Neoplasia — Practical Pathology

Outlines:

Identify the specimen → decide benign vs malignant → look for the diagnostic feature

Benign epithelial lesions: papilloma, viral wart, colonic polyp

Benign mesenchymal tumors: lipoma and leiomyoma

Progression to malignancy: dysplasia → carcinoma in situ → invasion

Malignant morphology: squamous cell carcinoma, adenocarcinoma and anaplasia

Routes of spread: lymphatic, hematogenous and transcoelomic seeding

Benign Tumors

Characterized by circumscription, differentiation, and absence of invasion.

Squamous Papilloma — Gross Appearance

Exophytic papillary lesion of stratified squamous epithelium

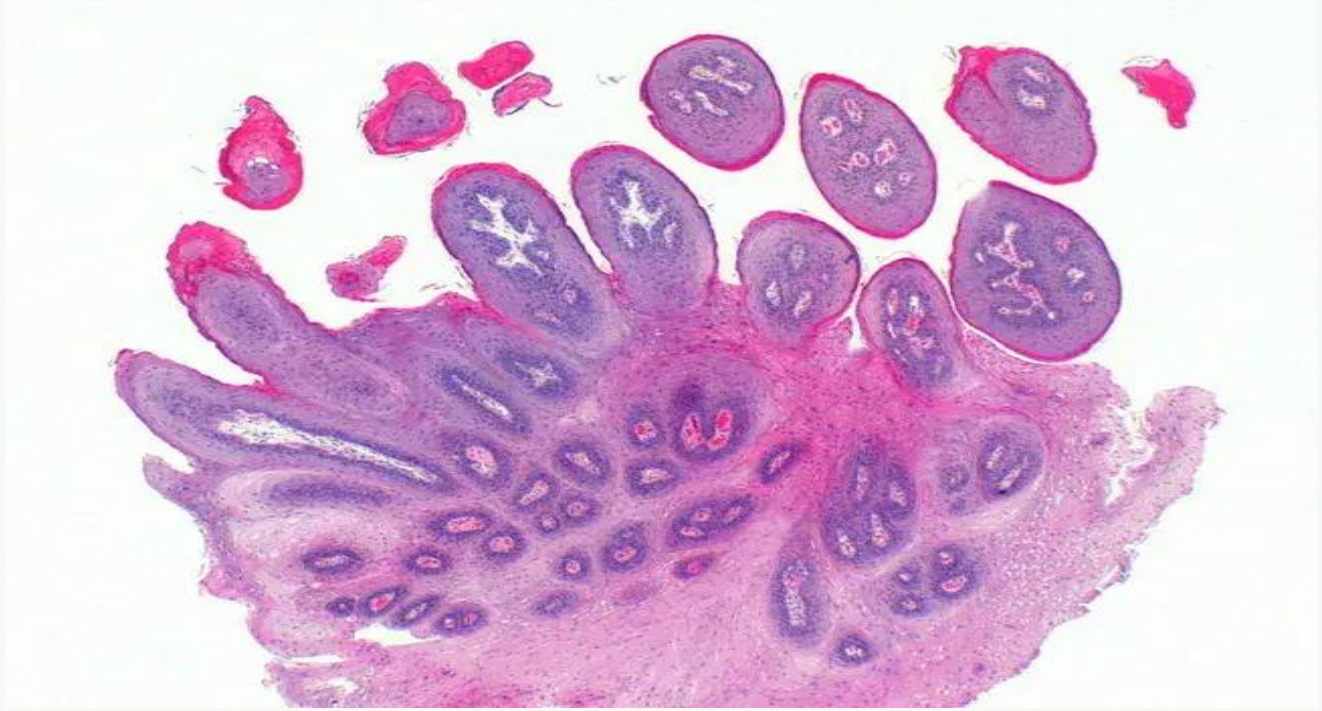




Oral squamous papilloma is a **benign, HPV-induced** proliferation of stratified squamous epithelium. **HPV types 6 and 11** are most common causative agents. It presents as a **small solitary, soft, painless exophytic mass** with numerous **finger-like projections.**

Squamous Papilloma — Microscopy

The hallmark is papillary fronds with a fibrovascular core





Squamous papilloma of the oral cavity is usually **pedunculated**. It is composed of numerous **finger-like papillary projections** lined by **keratinized stratified squamous epithelium** with **fibrovascular connective tissue cores**.

Viral Wart — Gross Appearance

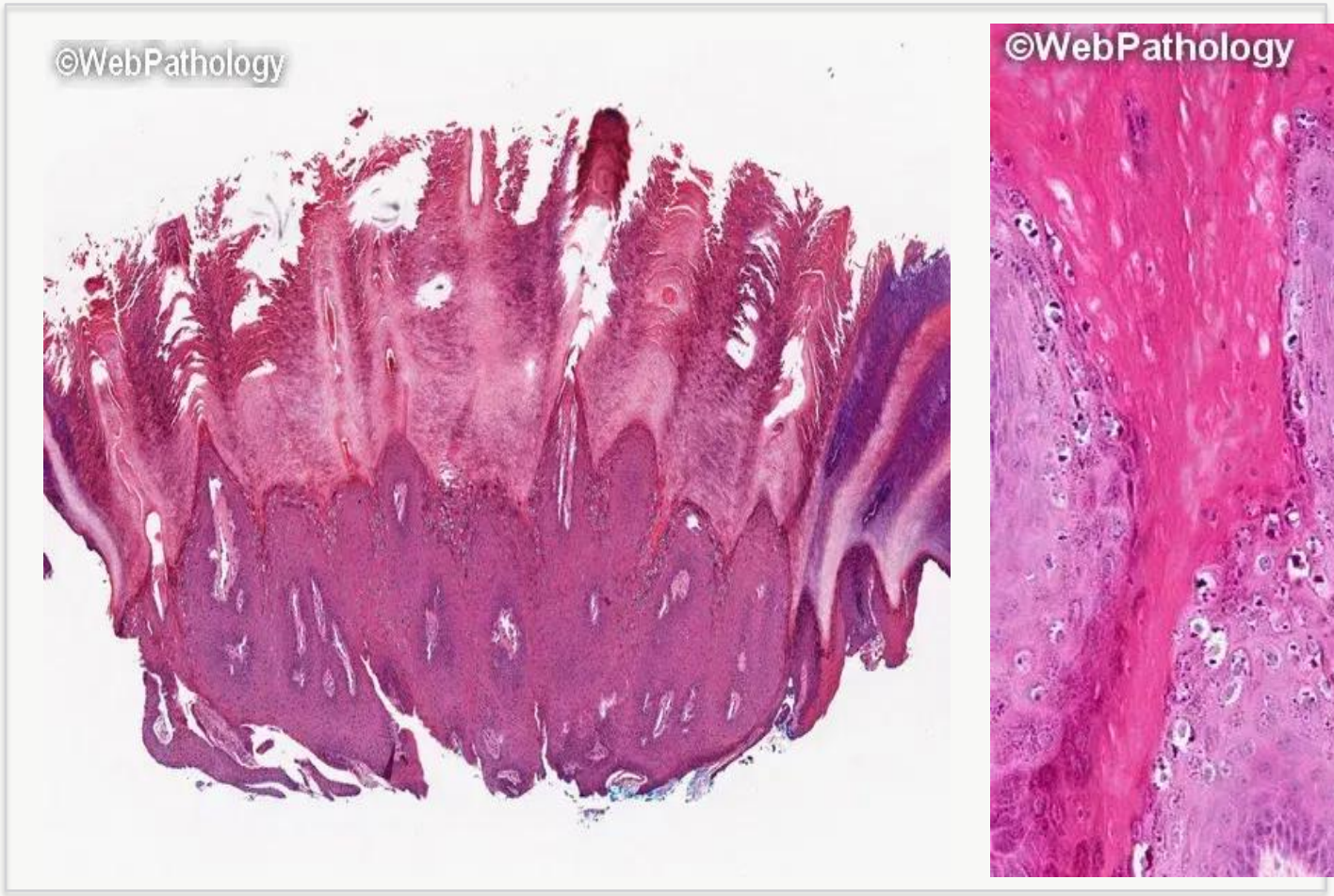
A rough, hyperkeratotic papule produced by HPV

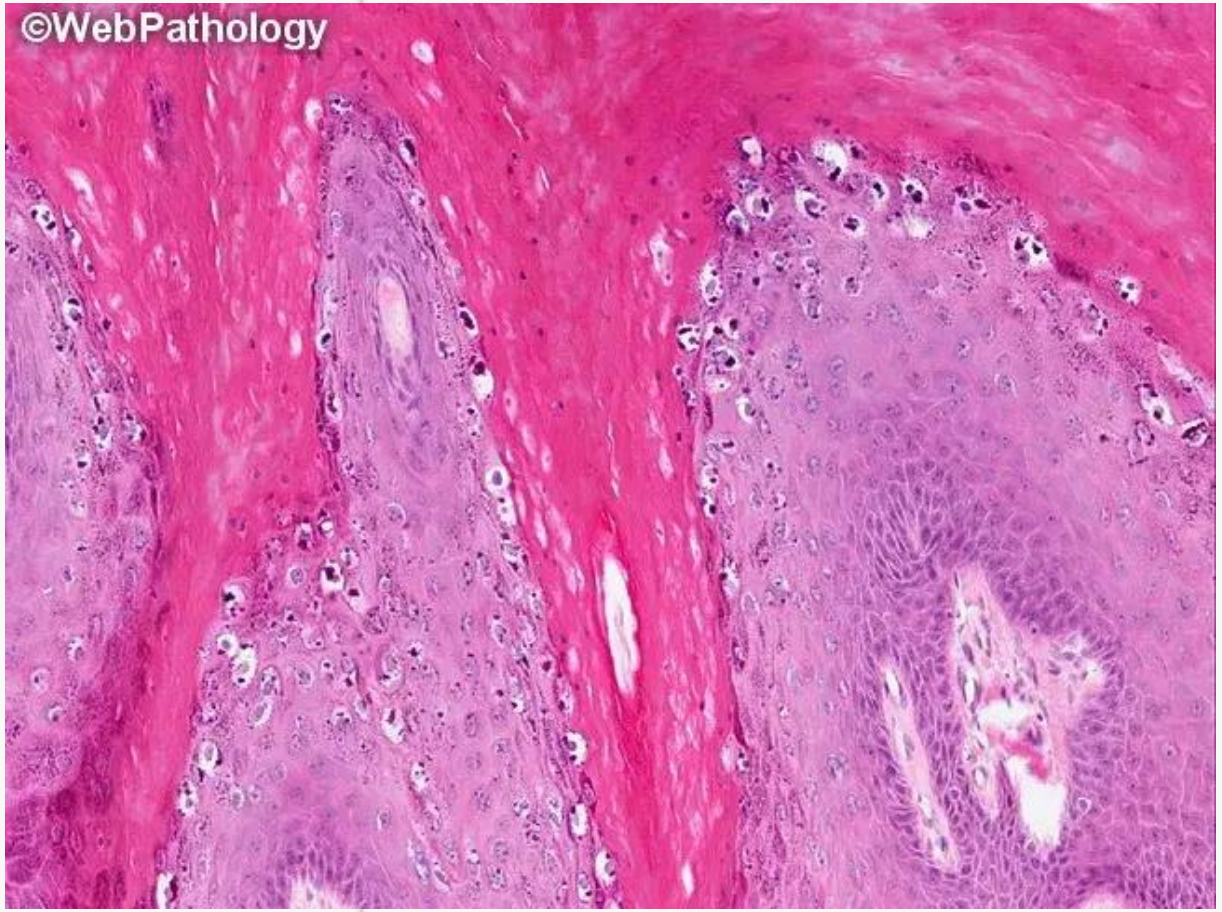


Plantar warts are usually only **slightly elevated** with rough or verrucous surface. They appears as **rough, callus-like thickenings** surrounded by hyperkeratotic skin. They are caused by **HPV.**

Viral Wart — Microscopy

Recognize hyperkeratosis, papillomatosis and koilocytosis

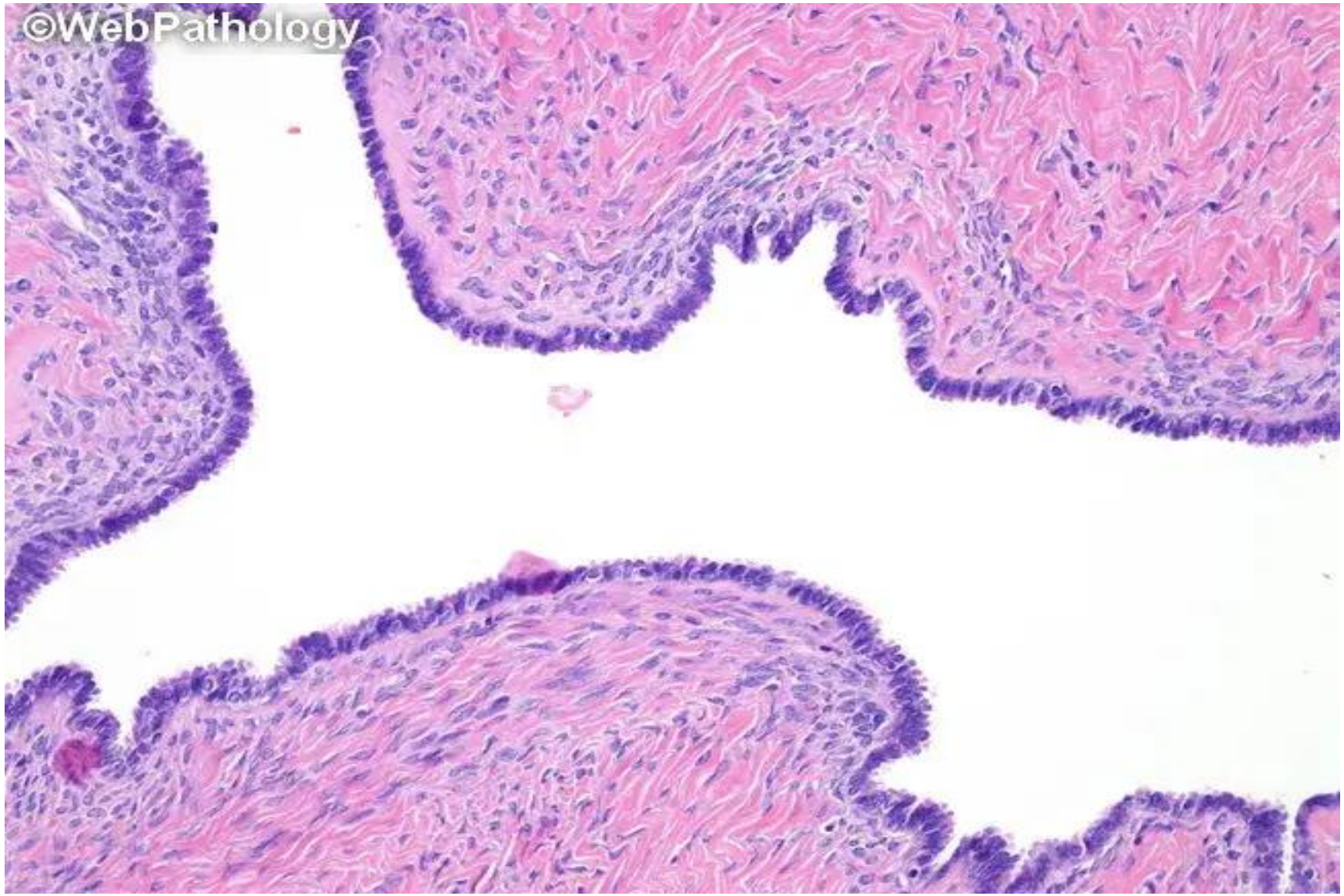




Hyperkeratosis, papillomatosis and hypergranulosis. **Koilocytes** are prominent in the upper layers of epidermis.



Serous cystadenoma of the ovary.



Serous cystadenomas of the ovary are **thin-walled unilocular cysts** that are **lined by ciliated pseudostratified cuboidal or columnar epithelium.**

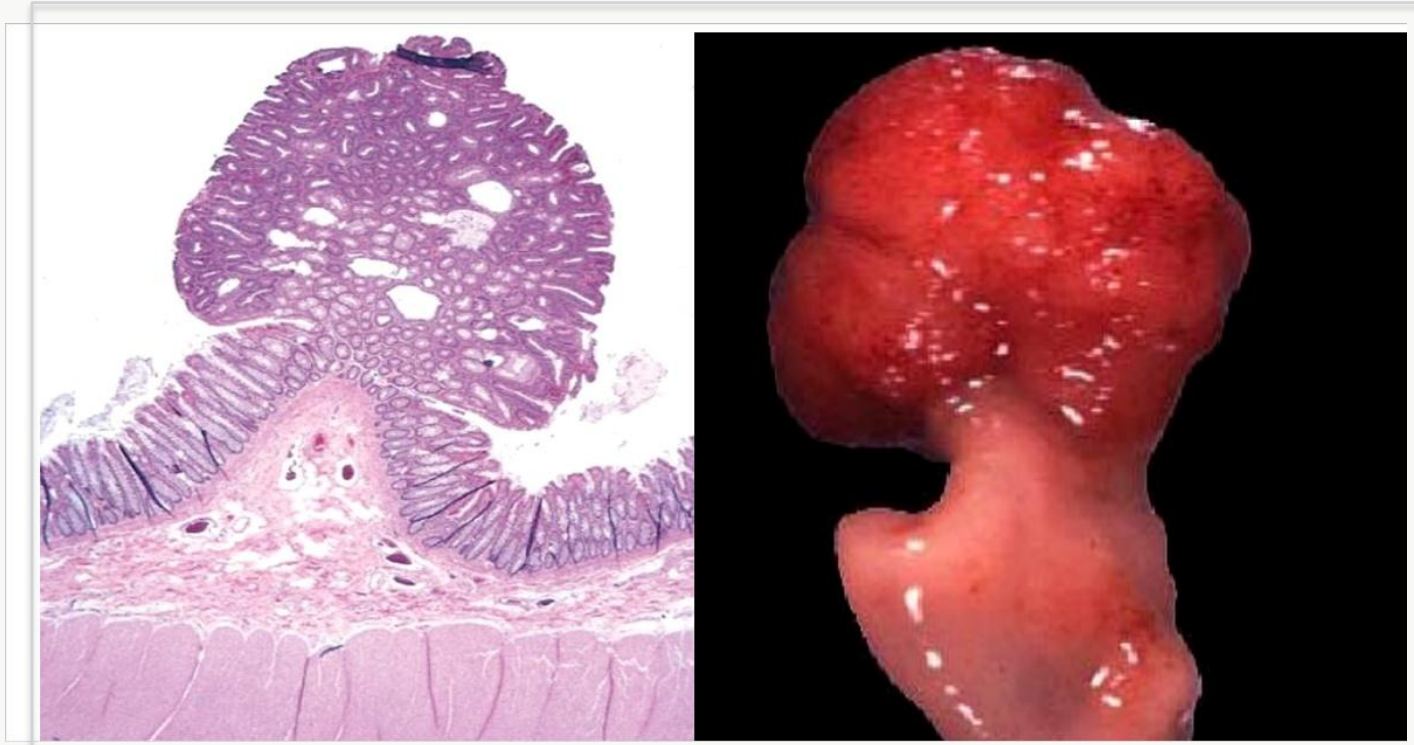




Papillary serous cystadenoma of the ovary. The tumor is **multicystic** and shows **fine papillary projections** from the cyst walls focally. The papillary projections, when present, are **lined by a single layer of cuboidal or columnar cells**

Colonic Polyp

Polyp: A mucosal projection supported by a fibrovascular core



Colonic polyp: Projects above the mucosal surface and covered by epithelial lining with fibrovascular stromal core

Benign Mesenchymal Tumors

Lipoma — Gross Appearance

A benign adipocytic tumor: yellow, soft and well circumscribed



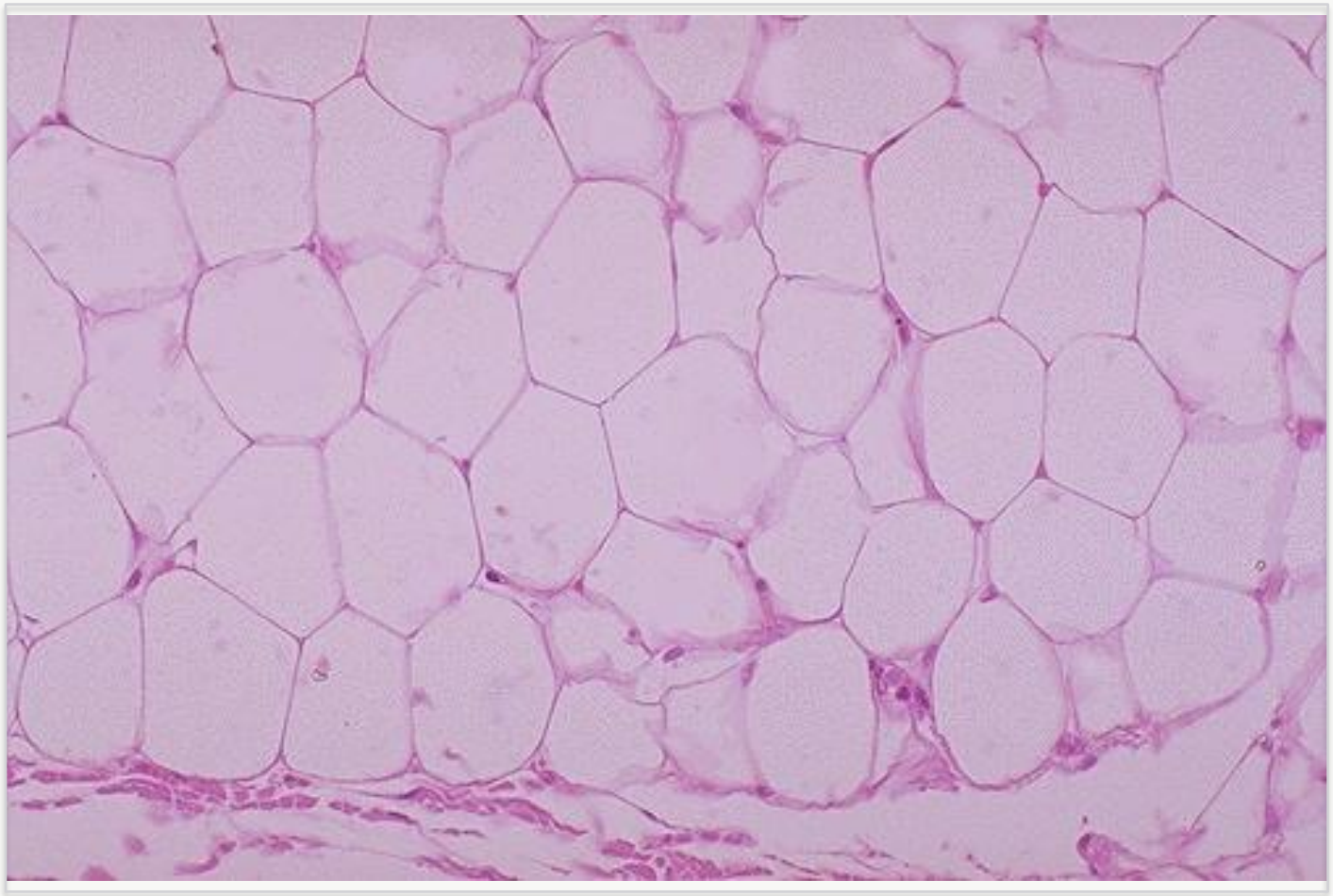
Well-circumscribed / often encapsulated

Soft to rubbery

Yellow cut surface reflecting mature fat

Lipoma — Microscopy

Uniform mature adipocytes without atypia



Mature adipocytes

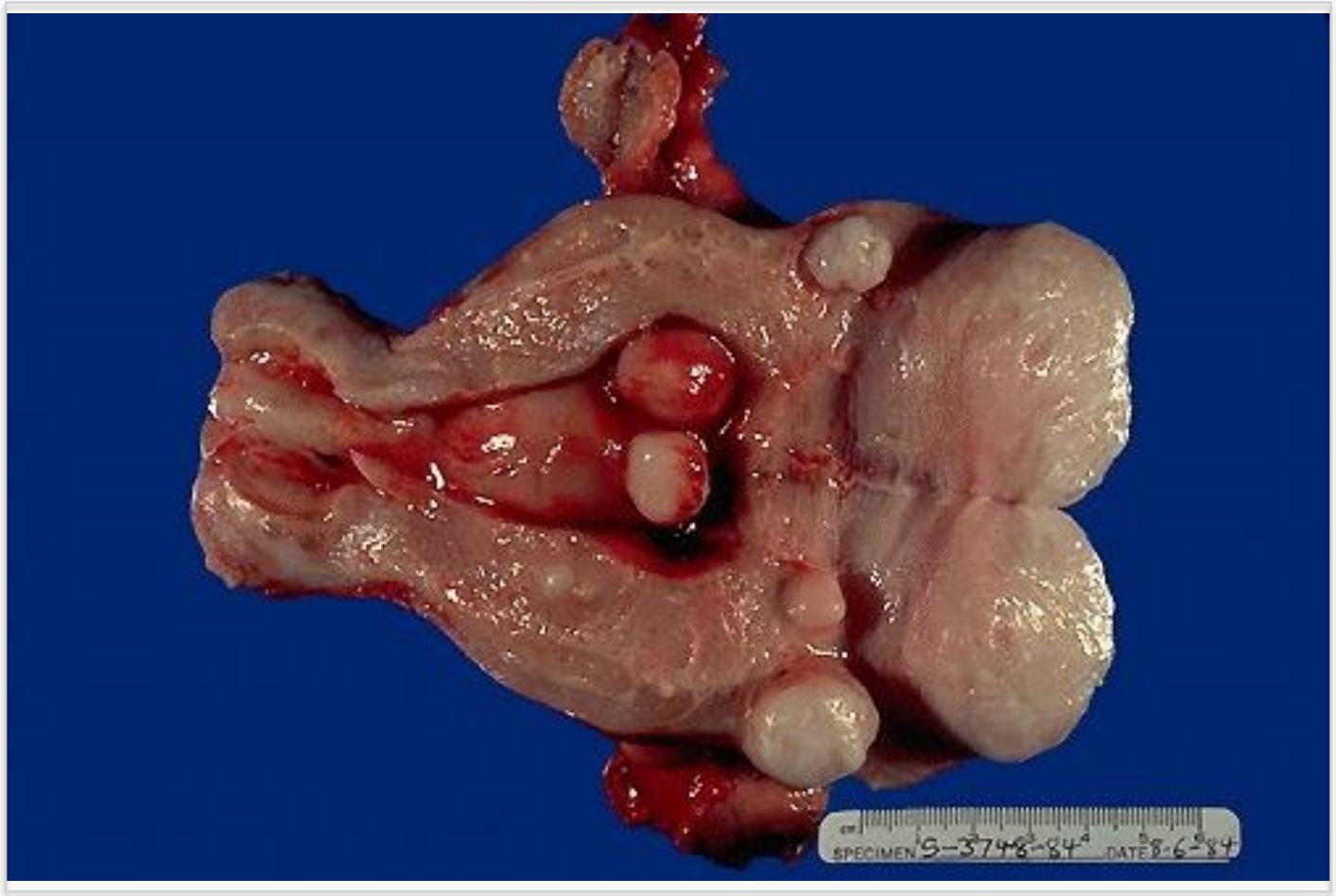
Uniform cell size

Peripheral small nuclei

No significant atypia or invasion to surrounding

Leiomyoma — Gross Appearance

A sharply circumscribed firm white mass with a whorled cut surface



Gross appearance of uterus with multiple leiomyomas

Well circumscribed

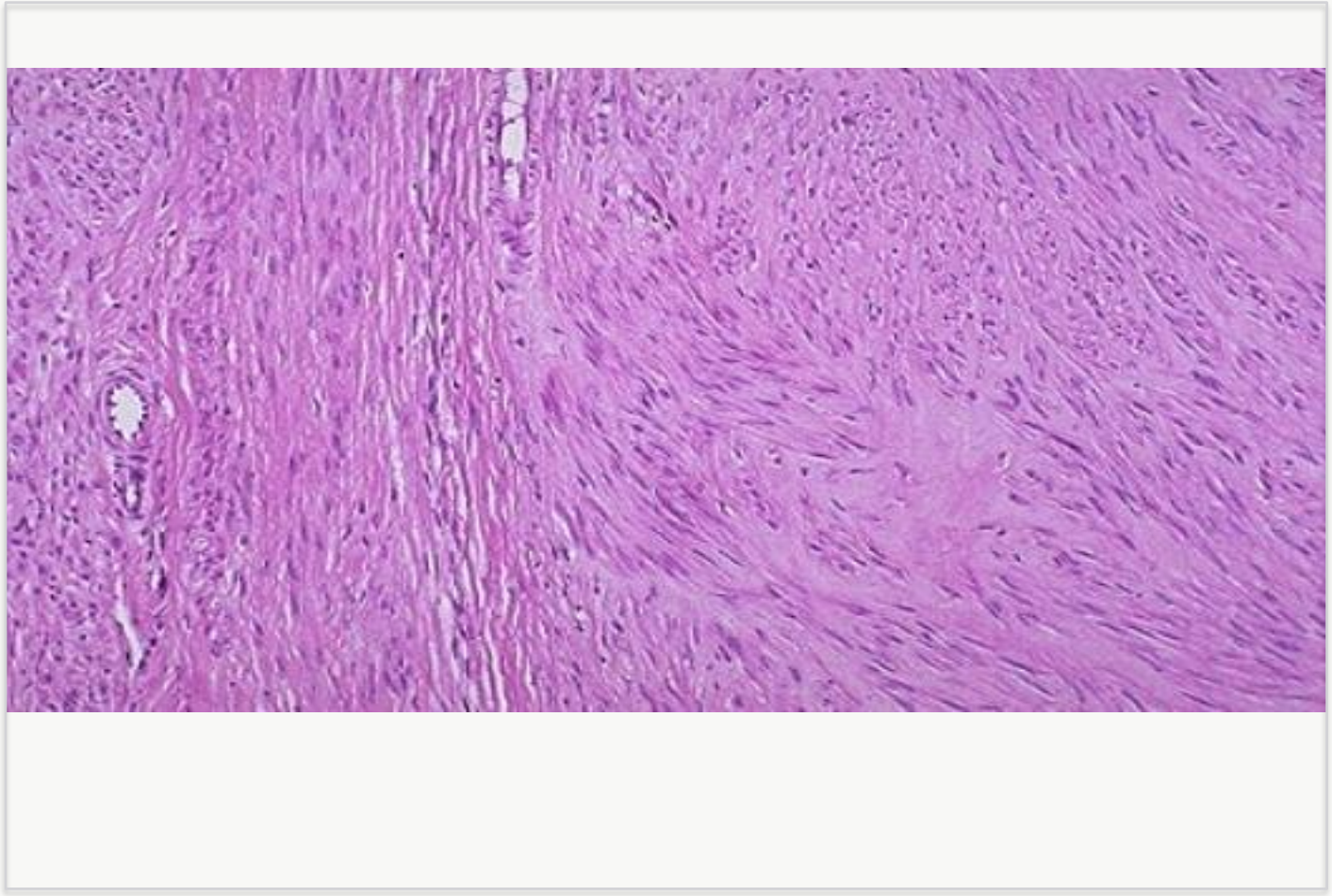
Firm, white-gray appearance

Classic whorled cut surface

Commonly multiple in the uterus

Leiomyoma — Microscopy

Interlacing fascicles of bland smooth-muscle cells



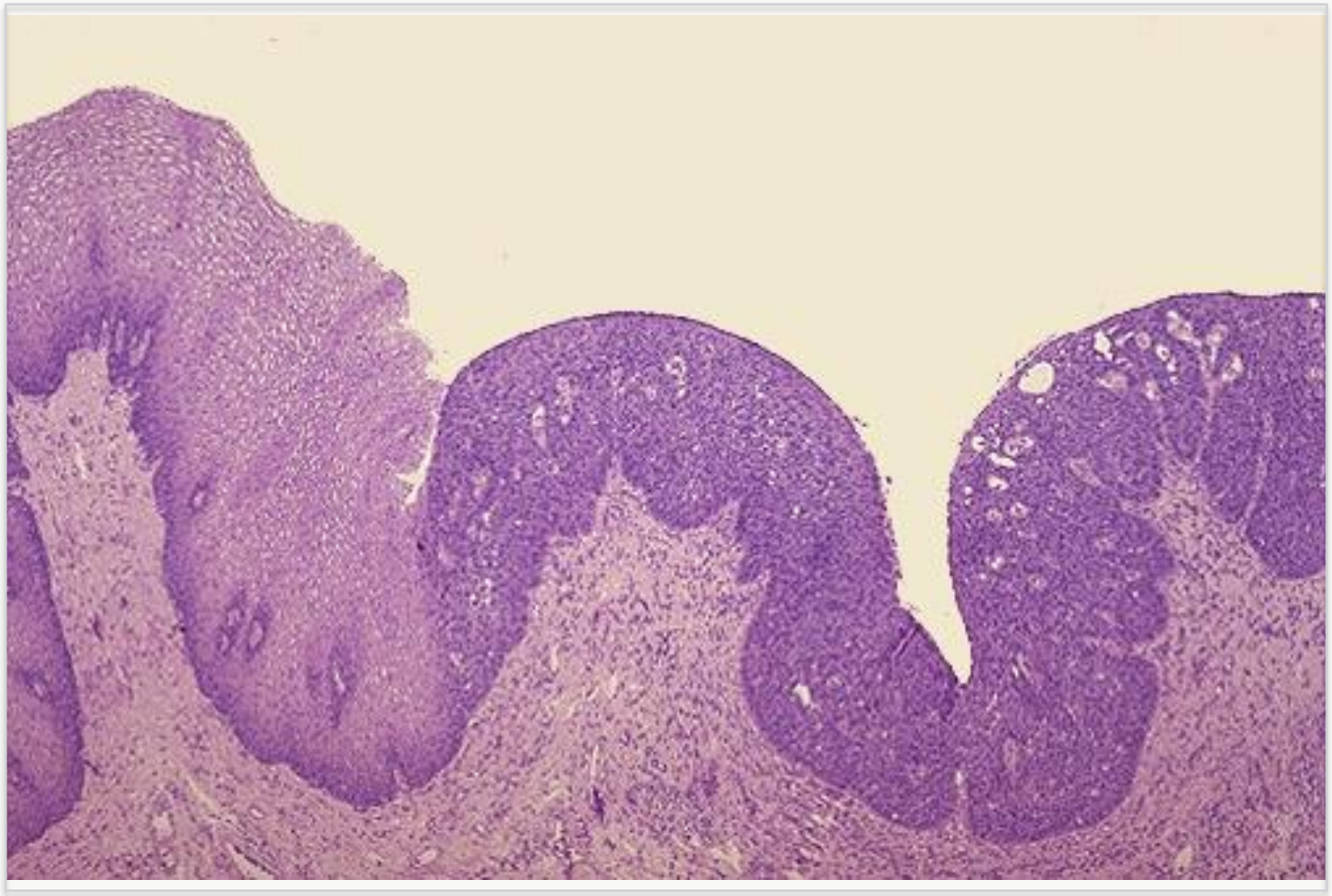
Densely packed spindle cells forming interlacing fascicles

The cells have cigar-shaped nuclei

Epithelial dysplasia

Epithelial Dysplasia

Atypical epithelial growth remains confined above the basement membrane



The normal epithelium adjacent to dysplasia

Disordered maturation

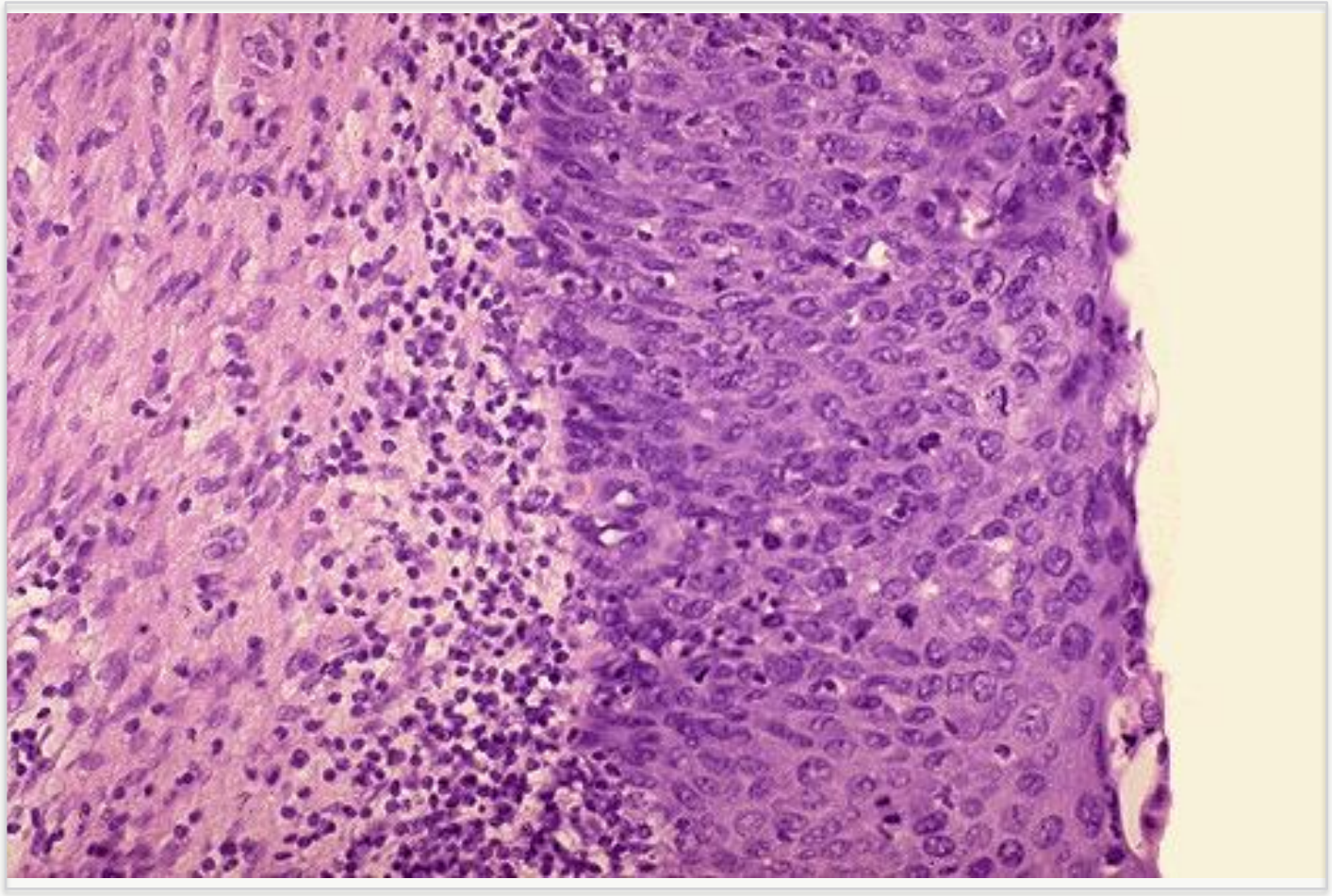
Cellular and nuclear atypia

Increased mitotic activity

Basement membrane remains intact

Carcinoma in Situ

Full-thickness epithelial dysplasia without stromal invasion



Atypia involves the full epithelial thickness

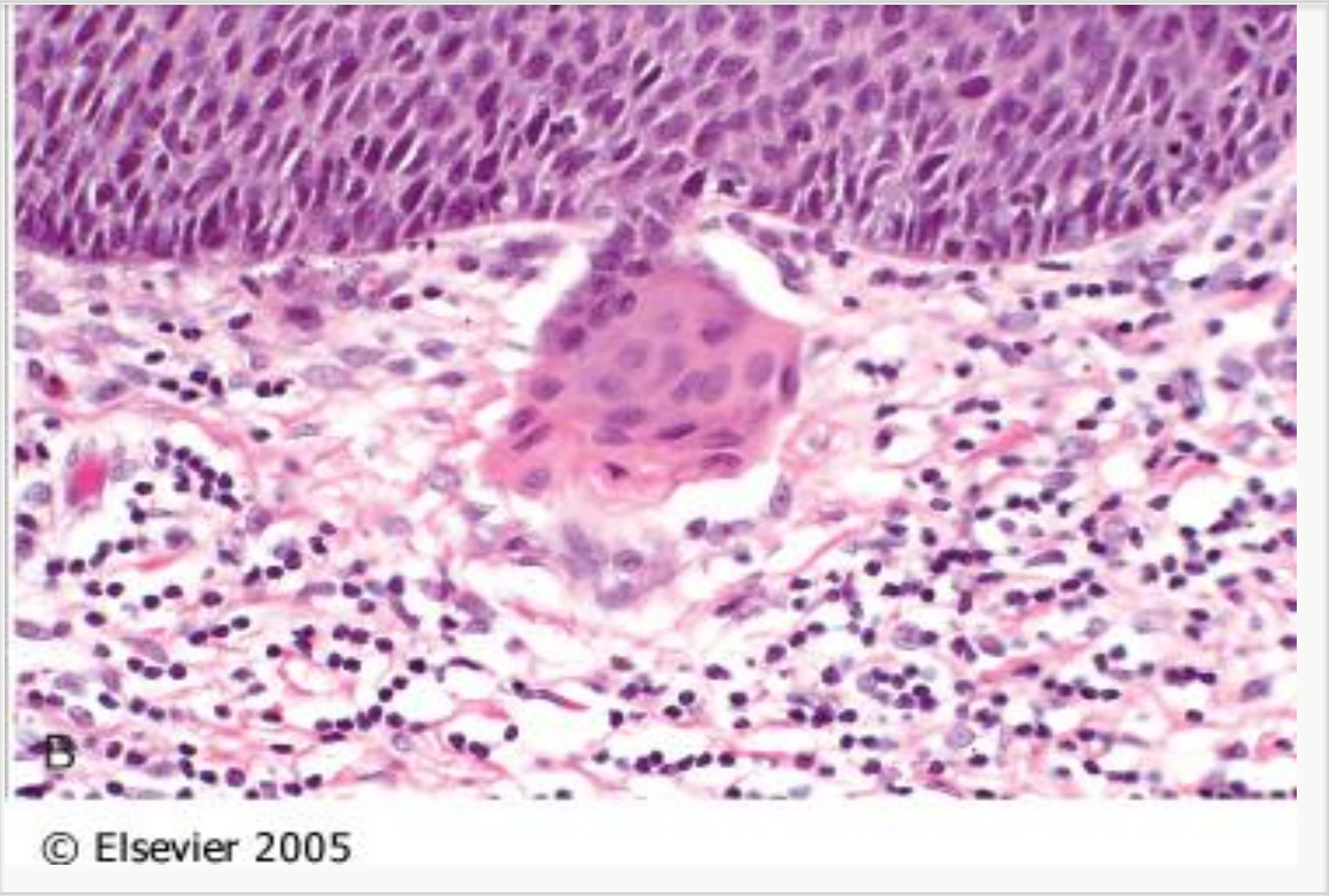
Loss of normal maturation

Basement membrane is intact

No invasion into underlying stroma

Invasive Carcinoma

Malignant cells breach the basement membrane and infiltrate stroma



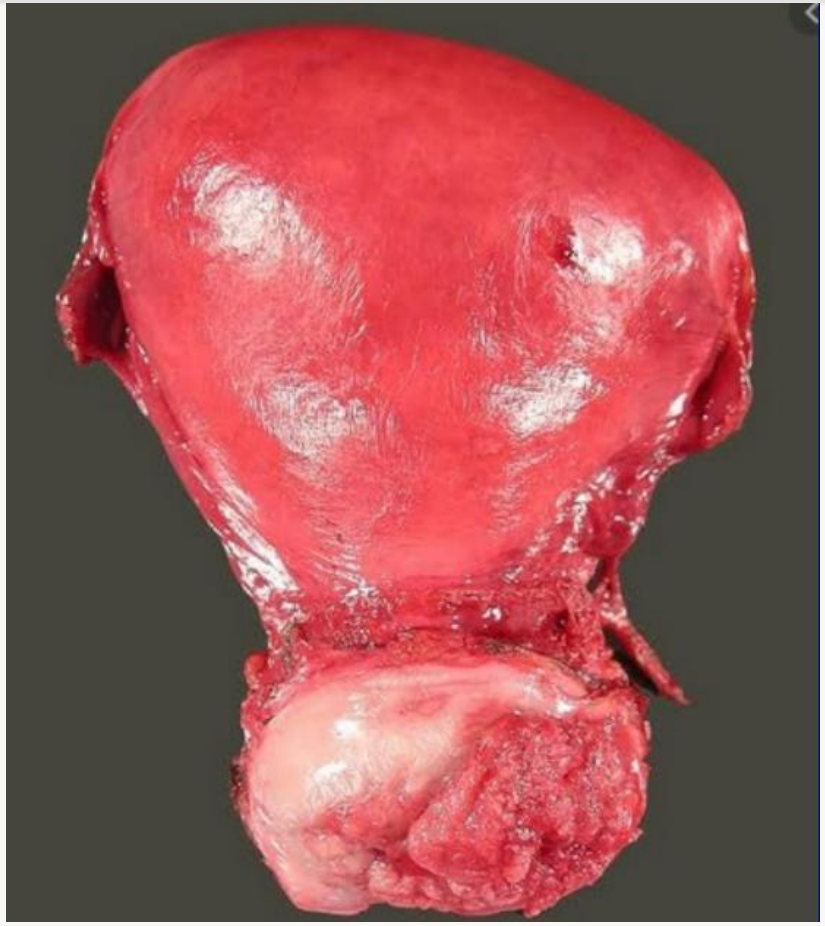
Cervical cancer with early stromal invasion

- Basement membrane is breached
- Tumor nests or cells extend into stroma
- Invasion distinguishes carcinoma from in situ

Malignant Tumors

Cervical Squamous Cell Carcinoma — Gross

An irregular friable mass replacing or distorting the cervix



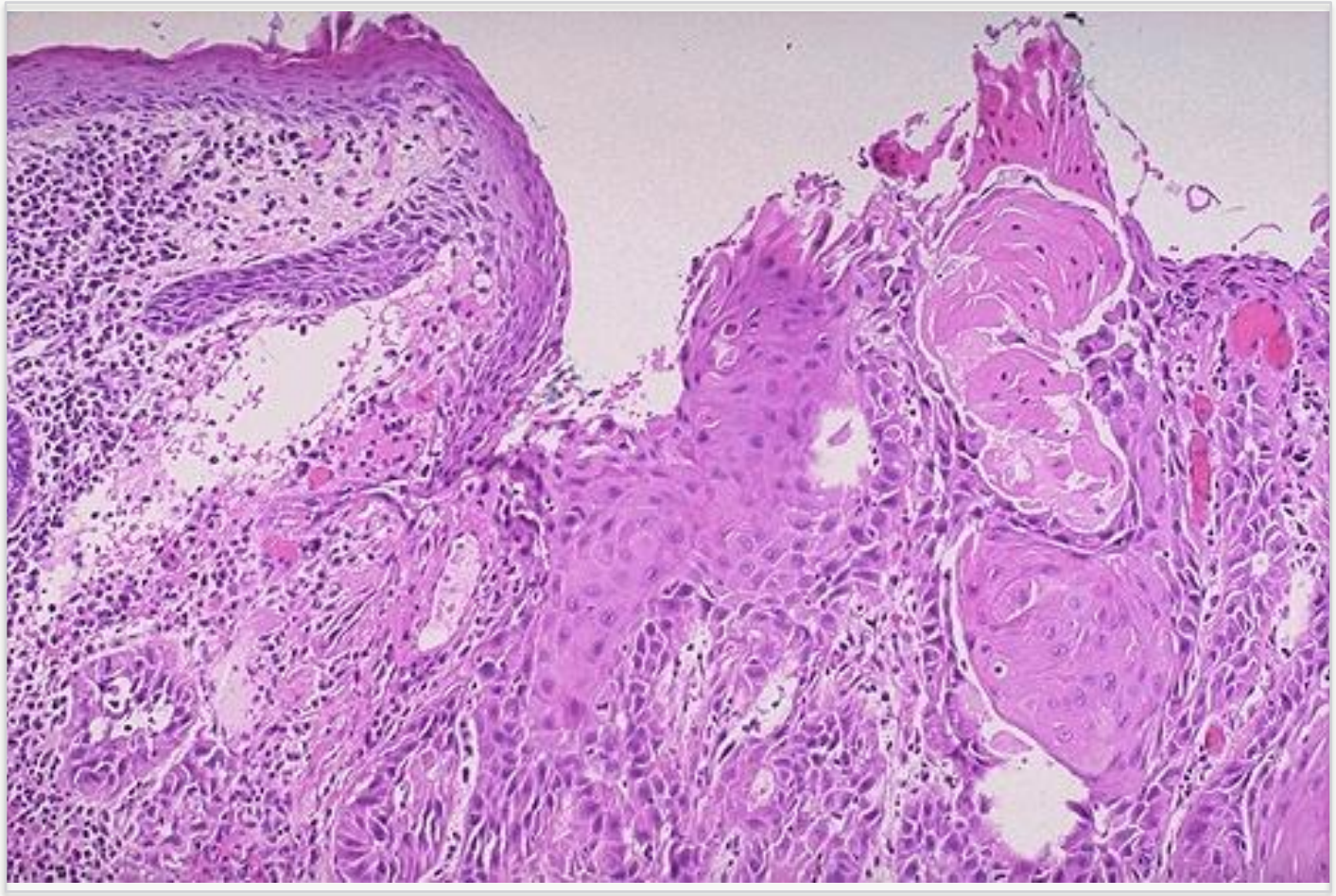
Irregular mass

Ulceration / friability may be present

Destructive growth rather than a smooth circumscribed border

Squamous Cell Carcinoma — Invasion

Well-differentiated squamous cells form invasive nests and cords



Downward stromal infiltration

Tumor cells resemble squamous epithelium

Architectural disorder and invasion confirm malignancy

Squamous Cell Carcinoma — Keratin Pearls

Keratinization is a classic clue to well-differentiated SCC



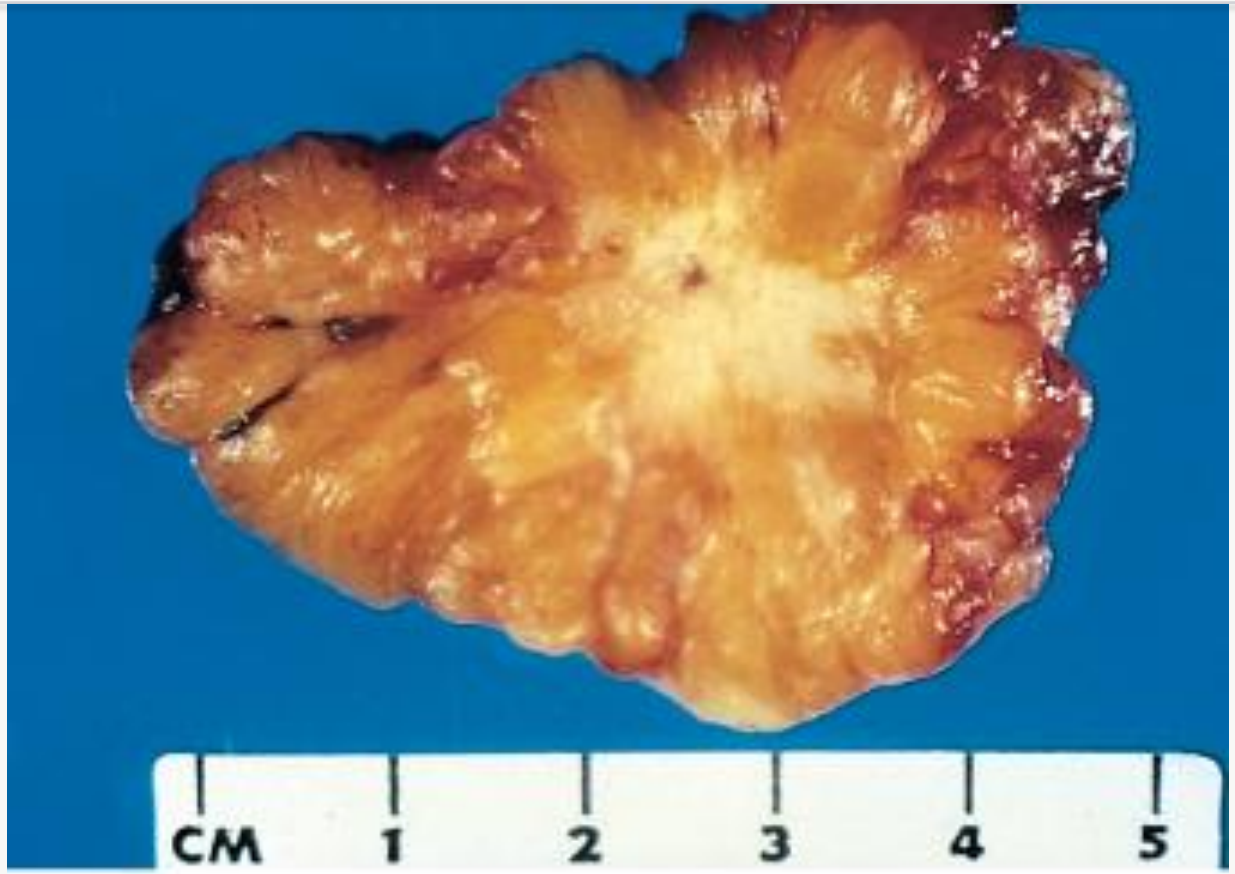
Nests of malignant squamous cells

Concentric eosinophilic keratin pearls

Intercellular bridges may be present

Invasive Carcinoma of the Breast — Gross

A hard irregular infiltrative mass with a stellate appearance

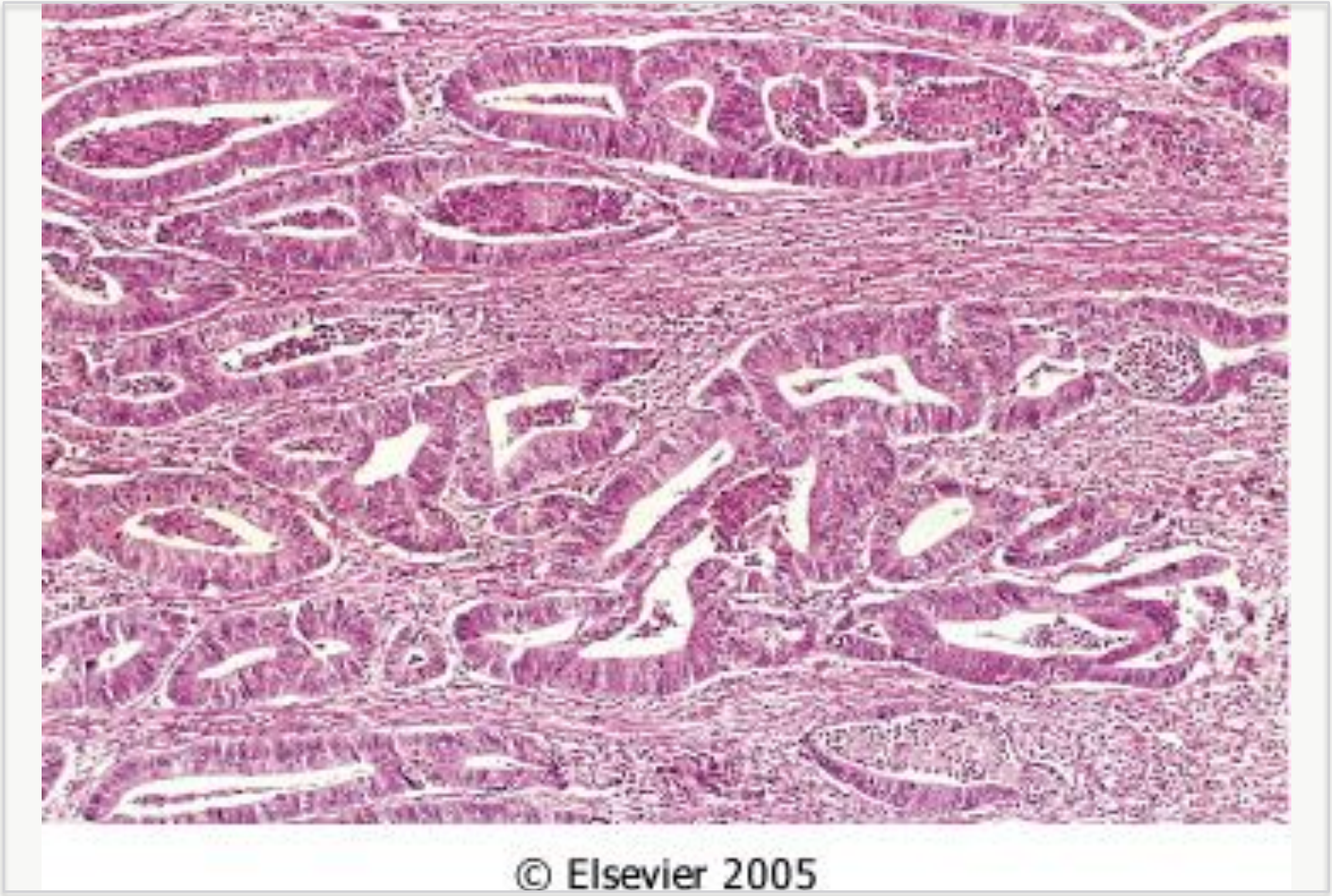


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- Poorly circumscribed with infiltrative borders
- Firm / stony hard
- Infiltrates surrounding breast tissue
- Often produces a stellate or scirrhous appearance

Invasive ductal carcinoma — Microscopy

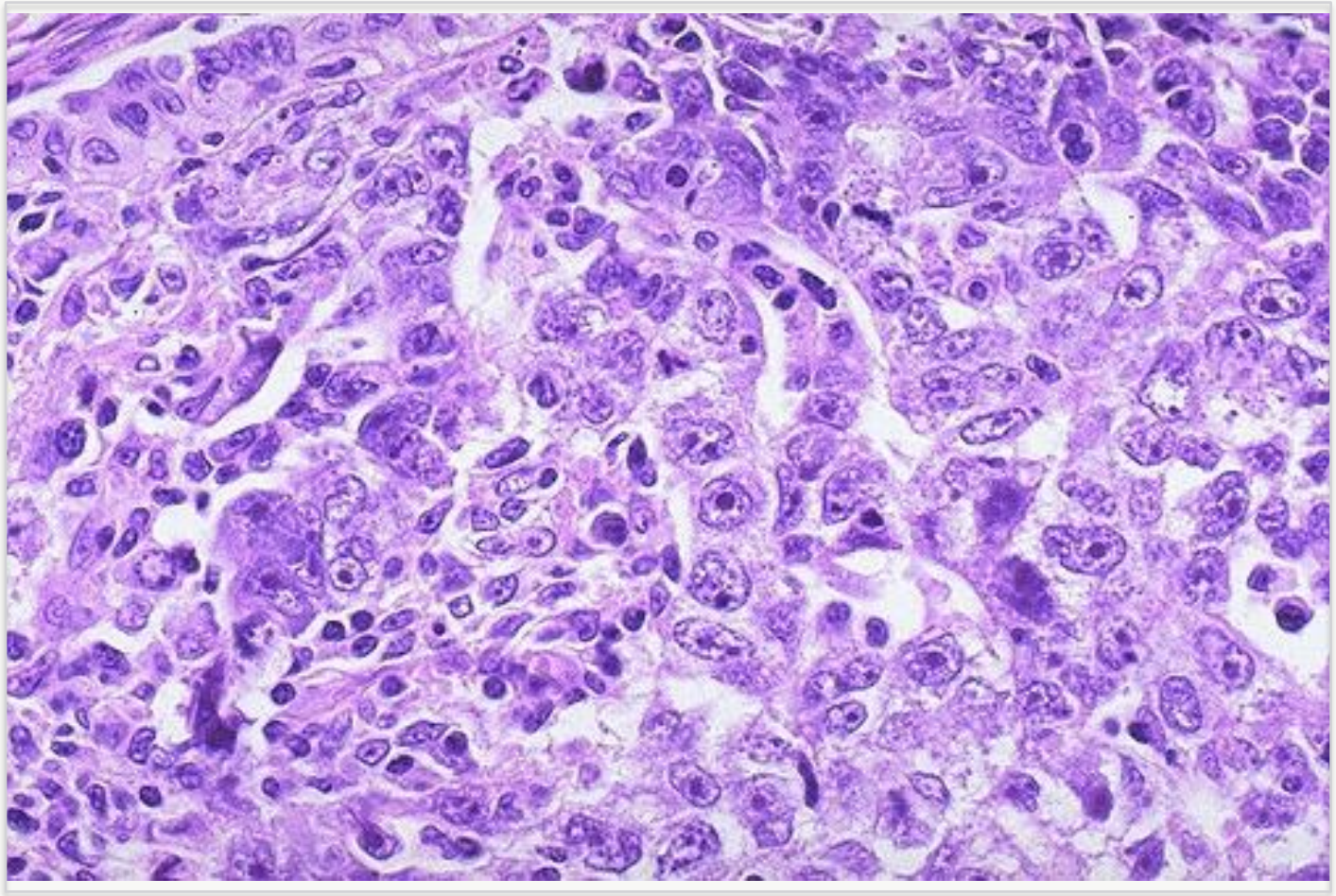
Malignant epithelial cells form infiltrative glandular structures



- Irregular malignant glands
- Invasion of supporting stroma
- Variable preservation of gland formation
- Degree of gland formation reflects differentiation

Anaplasia- poorly differentiated tumors

Loss of differentiation is accompanied by marked cytologic atypia



Poorly differentiated / anaplastic tumor

Marked pleomorphism

Hyperchromatic enlarged nuclei

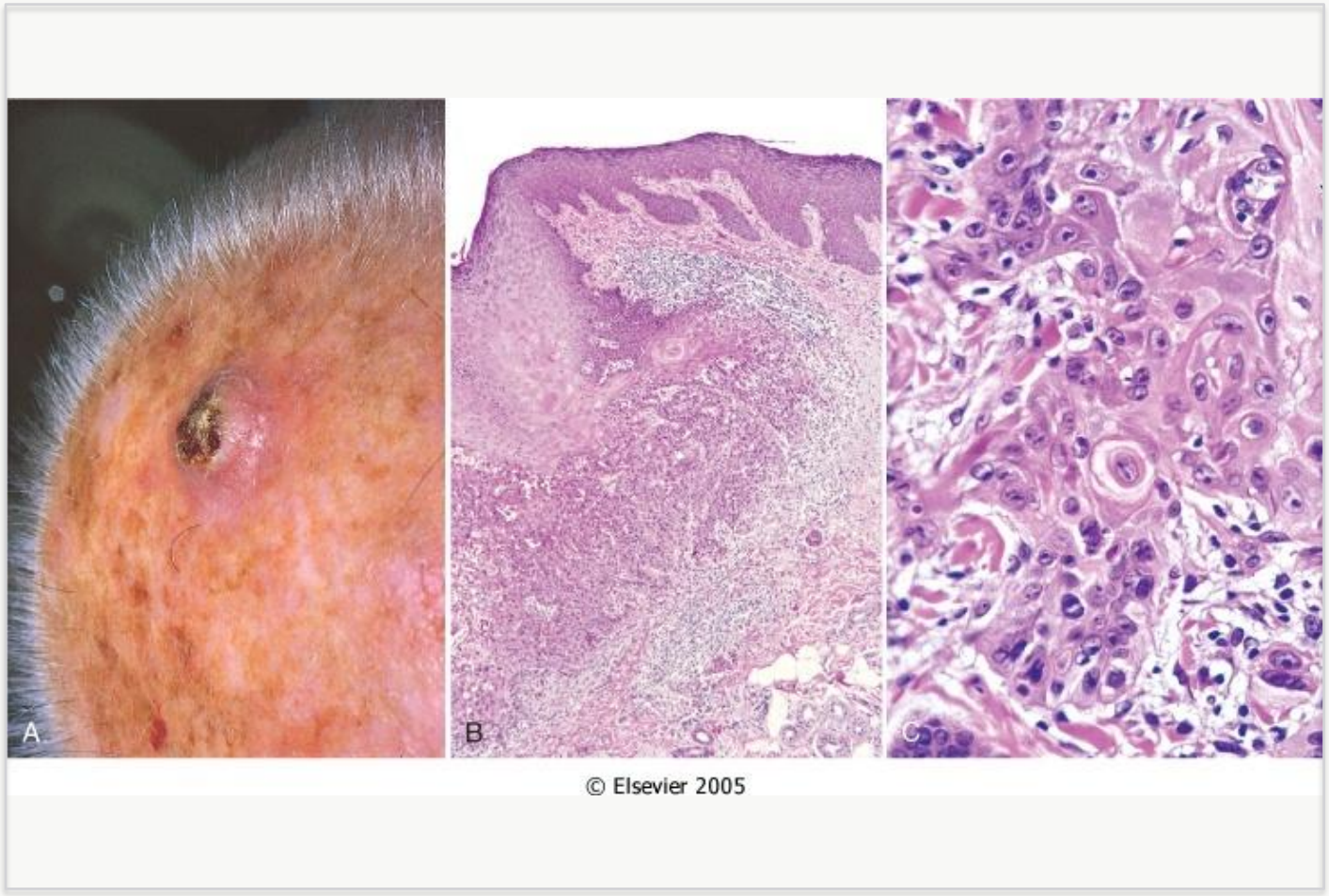
High nuclear-to-cytoplasmic ratio

Atypical mitoses may be present

Cell of origin may be difficult to identify

Squamous Cell Carcinoma of Skin

An invasive squamous carcinoma linked to cumulative UV exposure

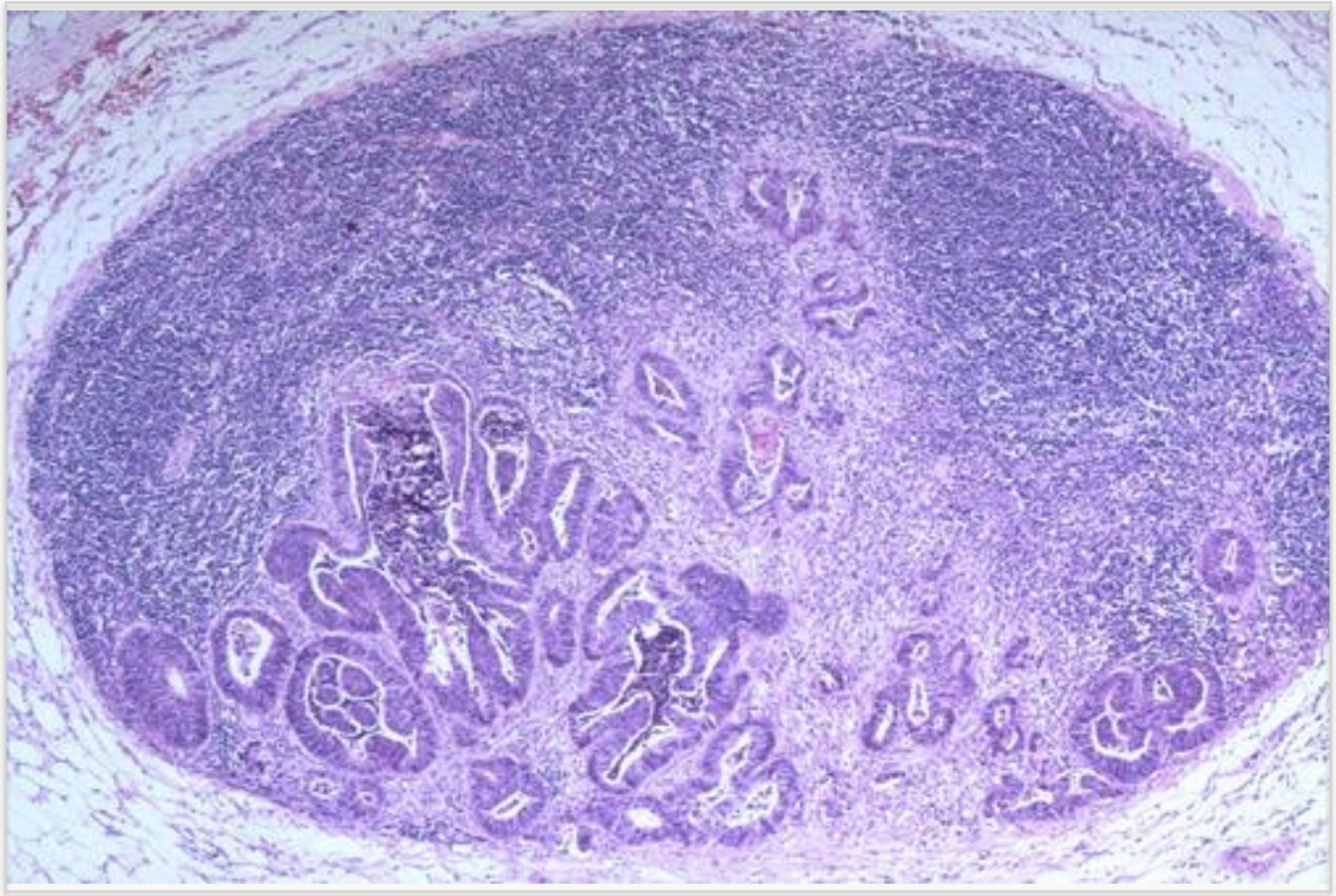


- Irregular cutaneous lesion
- Invasive nests of atypical squamous cells
- Keratinization may be visible microscopically
- UV radiation is an important risk factor

Tumor Spread & Metastasis

Lymphatic Spread

Carcinomas commonly spread first to regional lymph nodes



Microscopic lymph-node metastasis

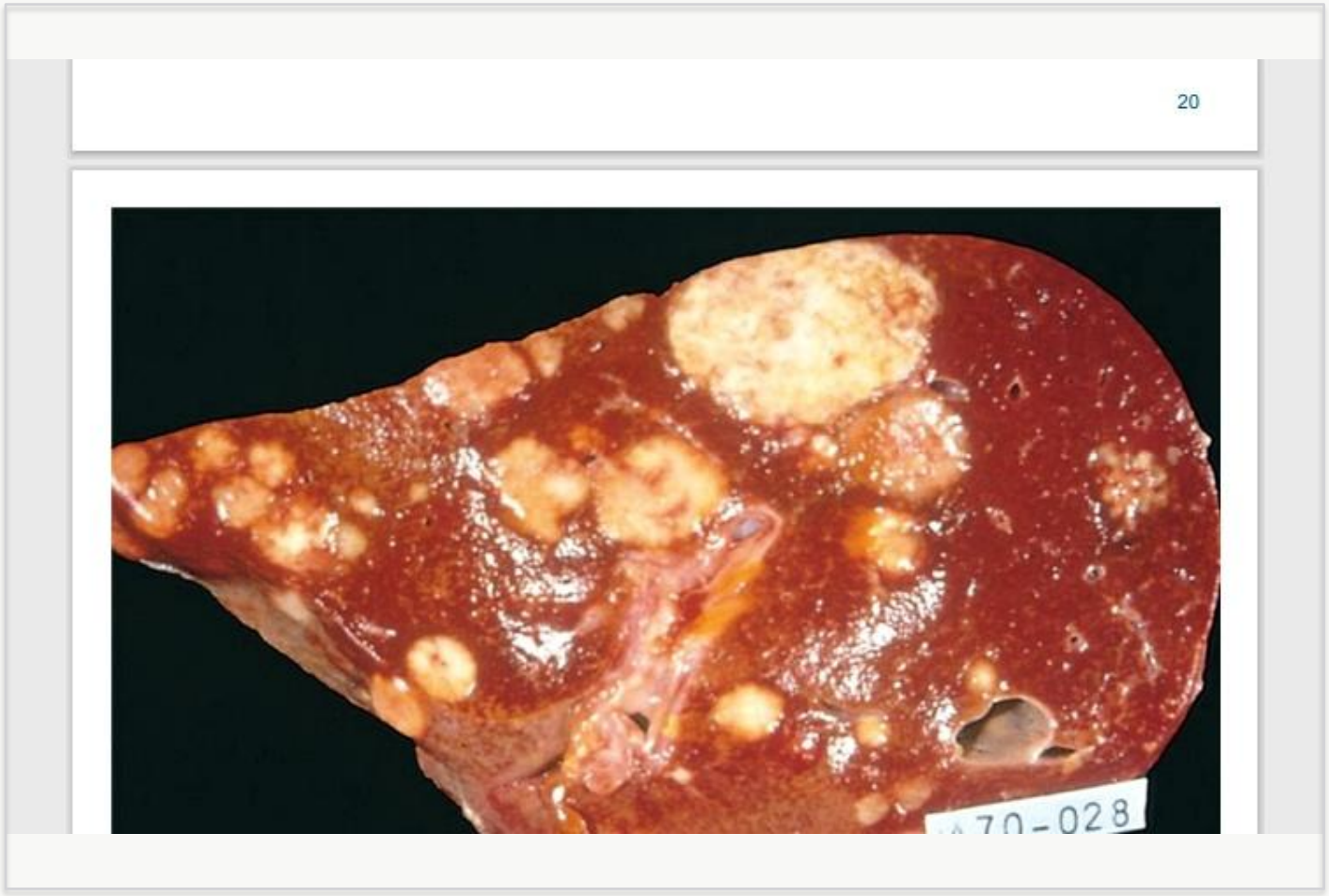
Metastatic tumor replaces part of a lymph node

Look for malignant glands or nests within nodal tissue

Lymphatic spread is common for carcinomas

Hematogenous Spread — Liver

Blood-borne metastases often produce multiple secondary nodules

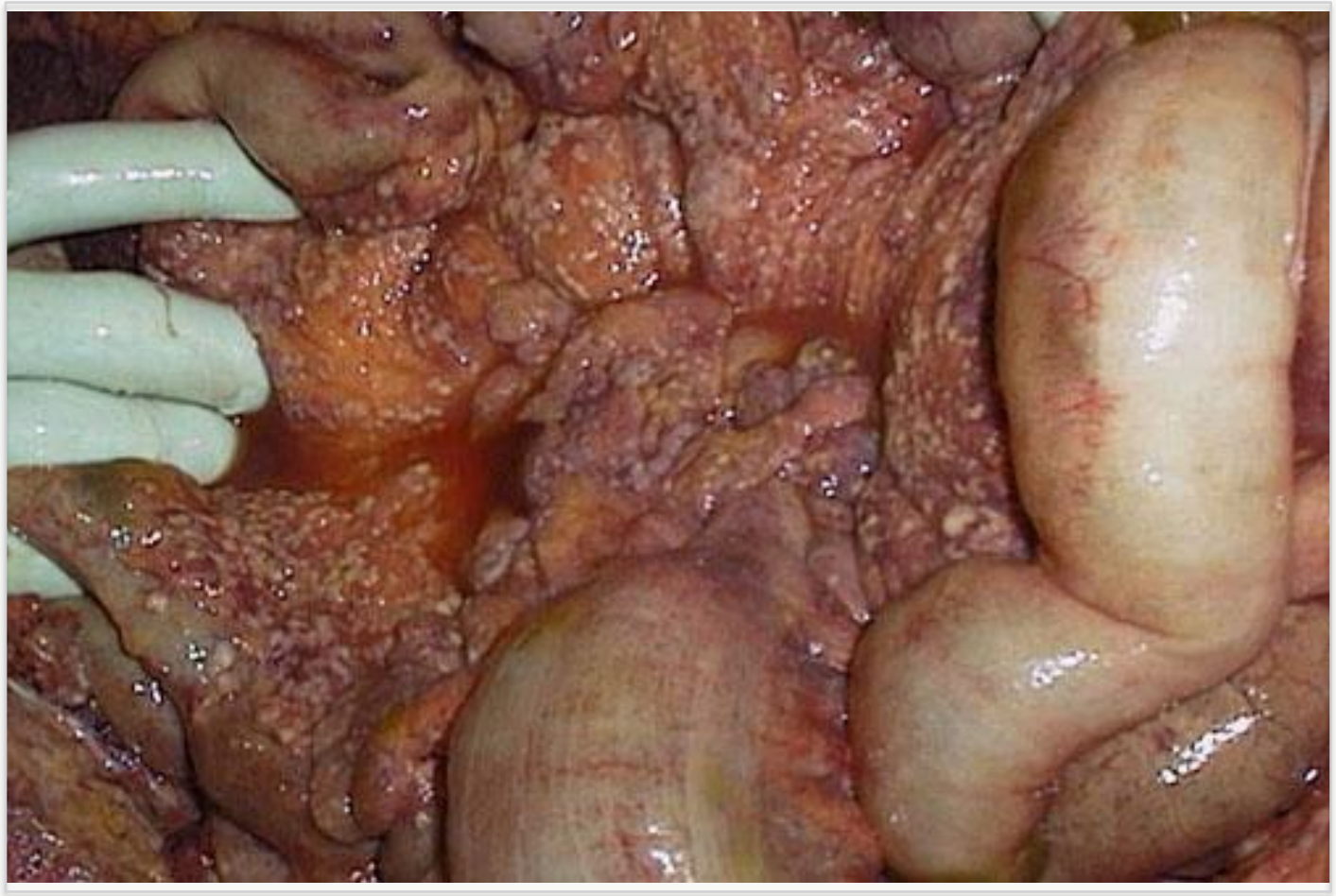


Gross: liver metastases

- Multiple tan-white masses
- Variable nodule size
- Sharp contrast with surrounding liver parenchyma
- Pattern favors metastatic disease over a solitary primary

Transcoelomic Spread — Peritoneum

Seeding across a body cavity produces numerous surface implants

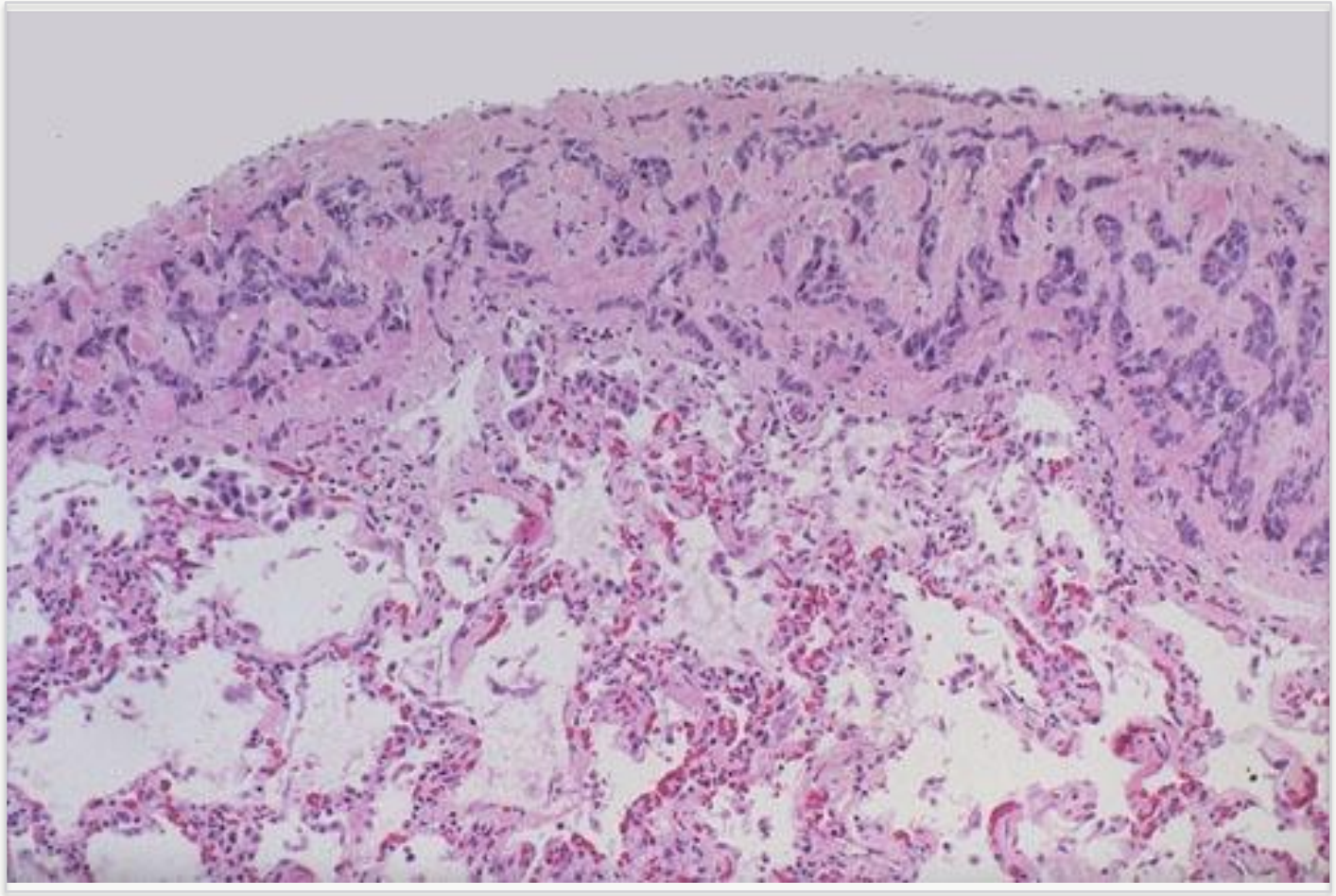


Peritoneal tumor seeding

- Numerous small tumor nodules
- Distributed across serosal surfaces
- Typical of transcoelomic spread

Pleural Metastasis

Tumor deposits can implant on pleural surfaces



Microscopic pleural metastasis

- Metastatic carcinoma involving pleura
- Represents spread to a body-cavity surface
- Correlate with known primary tumor

Practical Summary

Benign

- Well circumscribed
- Well differentiated
- Minimal atypia
- No invasion

Malignant

- Infiltrative border
- Variable differentiation
- Atypia / mitoses
- Tissue destruction

Metastatic

- Secondary deposits
- Often multiple
- May involve nodes, liver,
- pleura or peritoneum

Thank you